Research Proposal for Visiting PhD Program

Splitting Algorithms for Structured Nonconvex Nonsmooth Composite Optimization

Personal Information

Applicant and Program Information			
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Host Department:	Faculty of Mathematics	**Host Supervisor:**	Prof. Radu Ioan Boţ
Duration:	12 months (09/2026–09/2027)		
Host Supervisor's Research Area: • Numerical Methods for Convex Optimization and Monotone Inclusions • Nonsmooth Nonconvex Optimization Algorithms • Applications to Machine Learning			

1 Introduction

We consider a particularly expressive class of models in modern optimization and machine learning, given by the composite form

$$\min_{\mathbf{x}\in\mathcal{X}} \Phi(\mathbf{x}) := f(\mathbf{x}) + h(F(\mathbf{x})), \tag{1}$$

where $\mathcal{X} \subseteq \mathbb{R}^n$ is a nonempty, convex and compact set, where $f : \mathbb{R}^n \to \mathbb{R}$ is a smooth function, $F : \mathbb{R}^n \to \mathbb{R}^m$ is a smooth nonlinear mapping, and $h : \mathbb{R}^m \to \mathbb{R}\cup\{+\infty\}$ is a proper, closed, nonsmooth, and not necessarily convex function.

The nonsmooth composite Problem (1) is widespread in machine learning and statistical learning, where f is a data-fidelity loss function and h serves as a nonconvex regularizer inducing desirable structures (e.g., sparsity) in the solution. Importantly, many nonconvex regularizers admit difference-of-convex decompositions [12, 27, 26], with applications in compressed sensing [19, 31] and sparse regression [11, 32]. When h is an indicator function of a convex set C, this yields a constrained optimization formulation. In penalty methods [2], taking $h(F(\mathbf{x})) = \rho\|(F(\mathbf{x}))_+\|_1$ with $(F(\mathbf{x}))_+ = \max\{F(\mathbf{x}), \mathbf{0}\}$ yields the exact penalty formulation for functional constraints $F(\mathbf{x}) \leq \mathbf{0}$. Another motivating example arises in deep hashing [16, 20], where nonconvex regularization $h(\mathbf{y}) = \||\mathbf{y}| - \mathbf{1}\|_1$ is imposed on network output $\mathbf{y} = F(\mathbf{x})$ to drive $F(\mathbf{x})$ toward the discrete binary space $\{-1, 1\}^m$.

The explicit composite structure motivates us to develop splitting algorithms that exploit the smoothness of f, F via gradients and the nonsmoothness of h via proximal operators. It is crucial to develop efficient

algorithms with rigorous global convergence guarantees for nonconvex and nonlinearly coupled settings to ensure their reliability in practical applications.

1.1 Related work

Nonsmooth compositional optimization. In nonsmooth compositional optimization, such as Problem (1), significant work has been done in the case where $h \circ F$ is either convex [34, 24] or weakly convex [10, 8, 9, 25, 14]. Specifically, prior works [10, 8, 9, 25] typically ensure weak convexity of $h \circ F$ by imposing convexity on h and smoothness on F. Recent work [14] assumes both h and F are weakly convex with h being component-wise non-decreasing, thereby ensuring the weak convexity of $h \circ F$. However, when h itself is nonconvex, as in our proposal, the composite function $h \circ F$ is not necessarily weakly convex, rendering these methods inapplicable. For the fully nonconvex case, [3] reformulates the problem as constrained optimization and establishes asymptotic convergence via a Lagrangian-based method. Under the assumption that F is linear, some studies [4, 5, 7, 13] have proposed full splitting algorithms and established global convergence guarantees in a single-loop framework. Recently, [30] considers a similar setting to Problem (1) with nonlinear F. Using a smoothing technique, they provide non-asymptotic convergence analysis, while [33] systematically analyzed scenarios where h is Lipschitz continuous or non-Lipschitz continuous, establishing non-asymptotic and subsequential convergence results. Nevertheless, efficient single-loop methods with global convergence analysis for fully nonconvex, nonlinearly coupled composite problems remain largely unexplored.

Structured difference-of-convex problems. DC programming is challenging due to the nonsmooth and nonconvex nature of the objective. The classic DC algorithm (DCA) [23] linearizes the concave component via subgradients and iteratively solves convex subproblems. Following this idea of constructing tractable convex majorants, many variants, including proximal, Bregman, and stochastic versions, have been developed with global convergence or complexity guarantees [26, 18, 28, 17, 29, 21, 27]. Additionally, splitting methods have been applied to DC optimization [1, 6]. However, in Problem (1), the composite term $h(F(\mathbf{x}))$ is not necessarily a DC function even when $h = h_1 - h_2$ is DC-structured, due to the nonlinear coupling through F, making existing DC methods inapplicable. Recent work has begun addressing this challenge: [15] establishes a double-loop algorithm with subsequential convergence via exact penalty methods, while [30] provides non-asymptotic analysis via prox-linear methods. However, single-loop methods with rigorous full-sequence convergence guarantees remain underdeveloped for such nonsmooth nonconvex coupling.

1.2 Research Objective

To address these gaps, this research develops efficient single-loop splitting algorithms with global convergence guarantees for Problem (1) under two structural settings: general nonconvex h with access to proximal operators, and DC-structured $h = h_1 - h_2$.

2 Prior Work

Our prior research addressed the challenges of nonsmooth coupling and nonconvexity in Problem (1), where h is nonconvex and coupled with a nonlinear mapping $F(\mathbf{x})$. We have developed two efficient single-loop splitting methods with rigorous theoretical guarantees and practical effectiveness, which provide a foundation for the proposed research.

(i) Primal-dual splitting for deep hashing with nonconvex quantization regularizers. To address the coupling between nonconvex quantization regularization and nonlinear network outputs in deep hashing,

we leveraged Fenchel conjugacy to reformulate the nonconvex problem into a convex dual formulation with closed-form updates. This approach achieves complexity guarantees and demonstrates strong empirical performance on multiple datasets (DualHash, under review).

(ii) Penalty-based single-loop method for DC optimization with nonconvex constraints. While single-loop methods have been extensively studied for unconstrained DC problems, their application to constrained DC optimization remains largely unexplored. We designed a penalty-based framework that exploits the proximal operators of each DC component and gradient evaluations for the smooth part, achieving $\mathcal{O}(\varepsilon^{-3})$ complexity via variance reduction techniques (MoSSP, under review).

Both works tackle key aspects of the central research question: how to exploit nonconvex structure (DC decomposition, conjugate duality) to develop efficient splitting algorithms with rigorous convergence guarantees for nonlinearly coupled problems.

3 Research Design

We plan to address Problem (1) through two settings: **Problem 1** considers general nonconvex h with computable proximal operators, while **Problem 2** exploits DC structure $h = h_1 - h_2$.

3.1 Problem 1: Nonconvex h with Computable Proximal Mapping

We focus on Problem (1), where h is proper and closed, possibly nonconvex and nonsmooth, with a computable proximal operator. While promising progress has been made on splitting methods for nonconvex h under linear coupling [4, 5, 7], we extend this line of research to the nonlinear coupling $h(F(\mathbf{x}))$.

Objective. Our goal is to design a novel single-loop splitting algorithm that evaluates the proximal mapping of h and first-order information of f and F independently. We aim to establish rigorous convergence guarantees: subsequential convergence to approximate stationary points, with full-sequence convergence under mild conditions (e.g., the Kurdyka-Łojasiewicz property).

Key Challenges. The nonlinear coupling $h(F(\mathbf{x}))$ introduces fundamental difficulties:

(i) Intractable proximal mapping computation: The proximal operator of the composition $h \circ F$ is generally intractable, even in the case where the function h is convex.

(ii) Stationarity characterization under nonlinear coupling: Constructing an approximate stationarity measure compatible with single-loop updates remains an open problem under the nonsmoothness of h and nonlinear coupling.

Methodology. We proceed through the following steps.

(i) Algorithm Design: A full splitting algorithm will be developed that evaluates the proximal mapping of h and gradients of f and F independently.

(ii) Convergence Analysis:

(a) Stationarity construction: A tractable approximate stationarity measure will be constructed for Problem (1).

(b) Subsequential convergence: We will design a potential function and establish its sufficient descent property. Based on this, subsequential convergence to approximate stationary points will be established, along with iteration complexity bounds. Furthermore, under standard constraint qualification

(CQ) conditions (e.g., Theorem 10.6 and Corollary 10.9 in [22]), this result can be strengthened to exact stationary points of Problem (1).

(c) Full-sequence convergence: Under the Kurdyka-Łojasiewicz (KL) property, the full sequence convergence will be established.

3.2 Problem 2: Nonsmooth h with DC structure

We consider Problem (1) where h admits a difference-of-convex (DC) decomposition $h = h_1 - h_2$, and both h_1 and h_2 have tractable proximal mapping. The work [6] studies the case with linear operators acting on each convex component and proposes a full splitting scheme, avoiding the linearization of either convex term. Building on this line of research, the aim is to extend the splitting algorithms to settings with nonlinear mappings.

Objective. We aim to design a single-loop splitting algorithm that evaluates gradients, proximal operators, and nonlinear mappings independently, and establish subsequential convergence to approximate stationary points, with full-sequence convergence under the KL property.

Key Challenges. DC structure combined with nonlinear coupling introduces the following difficulties.

(i) Inapplicability of standard DC algorithms: Even when $h = h_1 - h_2$ is DC-structured, the compositions $h_1(F(\mathbf{x}))$ and $h_2(F(\mathbf{x}))$ are generally nonconvex due to the nonlinear coupling F, rendering traditional DC algorithms inapplicable. Moreover, their proximal operators may be intractable, preventing direct splitting schemes.

(ii) Stationarity characterization under nonlinear coupling: The DC structure involves subdifferential differences $\partial h_1 - \partial h_2$, while nonlinear coupling introduces additional chain rule terms through $\nabla F(\mathbf{x})$. We need to construct a tractable approximate stationarity measure that accounts for both aspects.

Methodology. We proceed through the following steps.

(i) Algorithm Design: A single-loop splitting algorithm will be developed that performs proximal evaluations for each nonsmooth component of the DC term separately, while avoiding proximal steps for nonsmooth compositions involving nonlinear operators. As a result, each iteration reduces to efficiently solvable subproblems.

(ii) Convergence Analysis:

(a) Construction of an approximate stationarity measure: An approximate stationarity/KKT measure will be constructed for the **DC** composite term, and the corresponding residual will be defined via the **splitting variables** and their proximal optimality conditions.

(b) Subsequential and full-sequence convergence: By designing a suitable Lyapunov function and parameter settings ensuring sufficient decrease, subsequential convergence to approximate stationary points of Problem (1) will be established, together with iteration complexity bounds. Moreover, under the Kurdyka–Łojasiewicz (KL) property and mild assumptions, global convergence of the full sequence will be proved.

(c) Practical implementation and acceleration: Acceleration techniques (e.g., inertial steps and adaptive strategies) will be investigated to improve practical convergence while preserving the theoretical guarantees.

Table 1: Research timeline

Stage	Phase	Research Tasks
Domestic Preparation	Feb – Aug 2026	Complete manuscript revisions (MoSSP, DualHash); Study classical splitting algorithms and convex analysis; Review Prof. Boţ's work; Conduct a comprehensive literature review on splitting methods with nonconvex h and nonlinear coupling; Identify key innovations and research gaps.
Visiting Research	Sep 2026	Environment familiarization; Refine research directions with Prof. Boţ.
	Oct – Dec 2026	Refine literature review and identify research gaps; Strengthen research motivation; Establish convergence analysis for Problem 1.
	Jan – Mar 2027	Perform numerical experiments; Refine convergence analysis for Problem 1; Draft and revise Paper 1.
	Apr – Jun 2027	Finalize and submit Paper 1; Initiate discussions on Problem 2; Conduct literature review and establish motivation.
	Jul – Sep 2027	Develop convergence analysis for Problem 2; Draft Paper 2; Summarize the research results and experiences.
Post-visit Work	Sep 2027 onwards	Finalize Paper 2 for submission; Integrate research outcomes into doctoral dissertation.

4 Research Plan and Expected Outcomes

4.1 Annual Research Plan

Domestic Preparation Phase (February – August 2026) During the months before the visit, I will complete revisions of two manuscripts currently under review and strengthen my theoretical foundation in convex analysis and splitting algorithms. I will then conduct a comprehensive literature review on splitting methods for Problem (1) with nonconvex h and nonlinear coupling, identifying key innovations and research gaps in the existing literature.

Visiting Research Phase (September 2026 – September 2027) The visiting period will focus on two main objectives: completing and submitting the first journal paper on Problem 1 through collaborative refinement with Prof. Boţ, and developing novel splitting algorithms for Problem 2.

Work After Returning to China (September 2027 onwards) Upon returning, I will finalize the second paper for submission, integrate research outcomes into my doctoral dissertation, and, if feasible, extend the work to stochastic settings.

4.2 Expected Outcomes

Two high-quality journal/conference papers.

4.3 Time Schedule

The research schedule consists of three stages: domestic preparation, visiting research, and post-visit work. Table 1 provides a detailed timeline.

5 Feasibility Analysis

The proposed research is feasible, supported by my research capabilities, the complementary expertise of both supervisors, and strong institutional resources.

(i) Solid research capabilities of the applicant. I have established a solid foundation in nonconvex optimization theory through both coursework and research practice. My prior work on primal-dual splitting for deep hashing and single-loop DC algorithms demonstrates proficiency in nonconvex nonsmooth optimization algorithm design and theoretical analysis, numerical validation, and academic writing in English.

(ii) Complementary expertise between supervisors and institutional support. Prof. Radu Ioan Boţ is a leading expert in operator splitting, monotone inclusions, and nonsmooth optimization—his theoretical expertise is highly relevant to the convergence analysis challenges in splitting algorithms. His expertise is widely recognized in the academic community, as evidenced by his editorial roles and numerous highly-cited publications. Under the mentorship of Prof. Boţ, we will benefit from strategic guidance in algorithm design and rigorous convergence analysis. The Faculty of Mathematics at the University of Vienna is highly ranked internationally, providing access to world-class research resources and collaborators. My domestic supervisor, Prof. Chunfeng Cui, specializes in optimization algorithms and theory, ensuring the proposed methods maintain both theoretical rigor and practical relevance. Beihang University's research team, led by Dean Deren Han, provides strong support in nonconvex optimization, aligning with the theoretical foundations of this proposal. This proposal has been discussed with both supervisors, ensuring objectives are realistic and mutually reinforcing.

(iii) Research continuity ensures timely PhD completion. The proposed research directly contributes to my doctoral dissertation on splitting algorithms for nonconvex composite optimization. Building on my prior work on primal-dual methods and DC algorithms, this visiting research advances the nonlinearly coupled component of my thesis. The planned two papers will form core chapters of my dissertation, ensuring both research depth and timely completion of my PhD program.

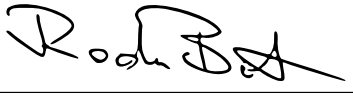
6 Work after returning to China

(i) Summarize the outcome of studying abroad and finalize related English papers.

(ii) Continue with further study and writing my doctoral dissertation based on the previous outcomes.

(iii) Bring advanced theories, methods, and techniques from the host research group to the domestic group, facilitating long-term research cooperation between both groups.

Signatures

SIGNATURE OF HOST SUPERVISOR: 

Date: 2026/02/09

SIGNATURE OF DOMESTIC SUPERVISOR: Chunfeng Cui

Date: 2026/02/09

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